

Geospatial Data Analysis

Workora Data Analyst Internship — Project Report

Dataset note

The task page did not provide a downloadable sample dataset. A representative self-created dataset of 30 Indian cities was therefore used to demonstrate the required geospatial analysis workflow. The values in this report should not be interpreted as official Workora or real-world business statistics.

Objectives

- Clean and validate geographic and business data.
- Calculate user penetration, revenue per user, and user density.
- Compare revenue and user distribution across states.
- Identify high-population cities with comparatively low user penetration.
- Create an interactive geographic visualization for exploration.

Data Preparation

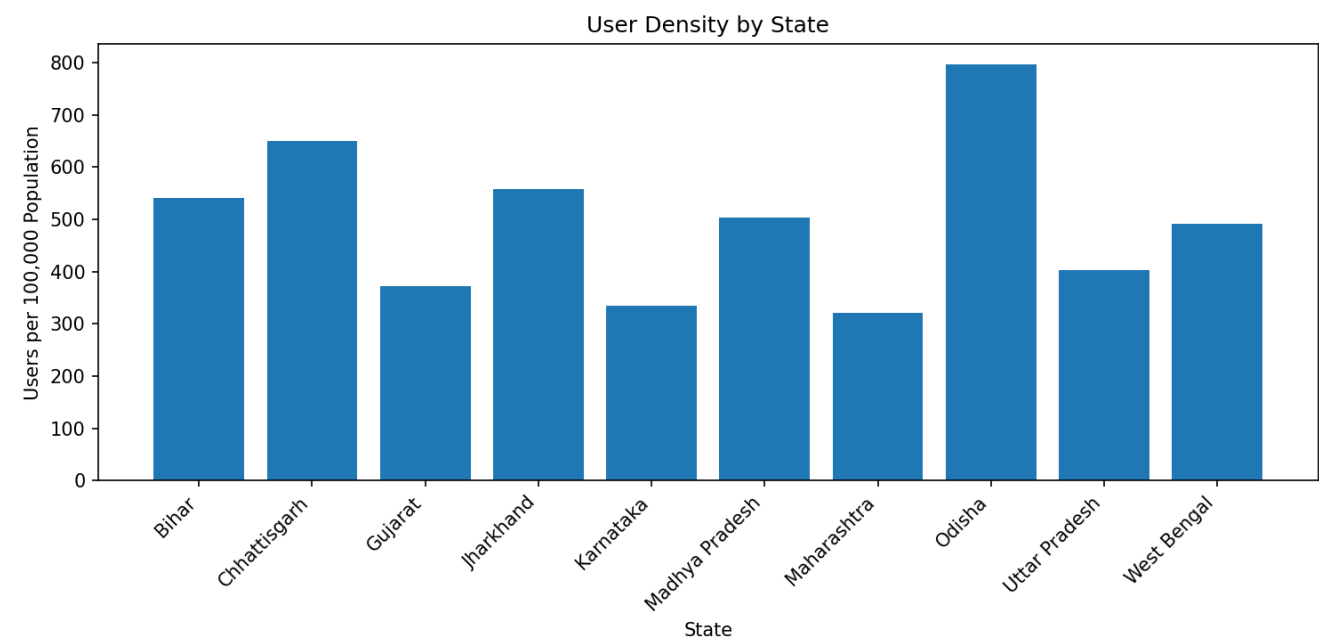
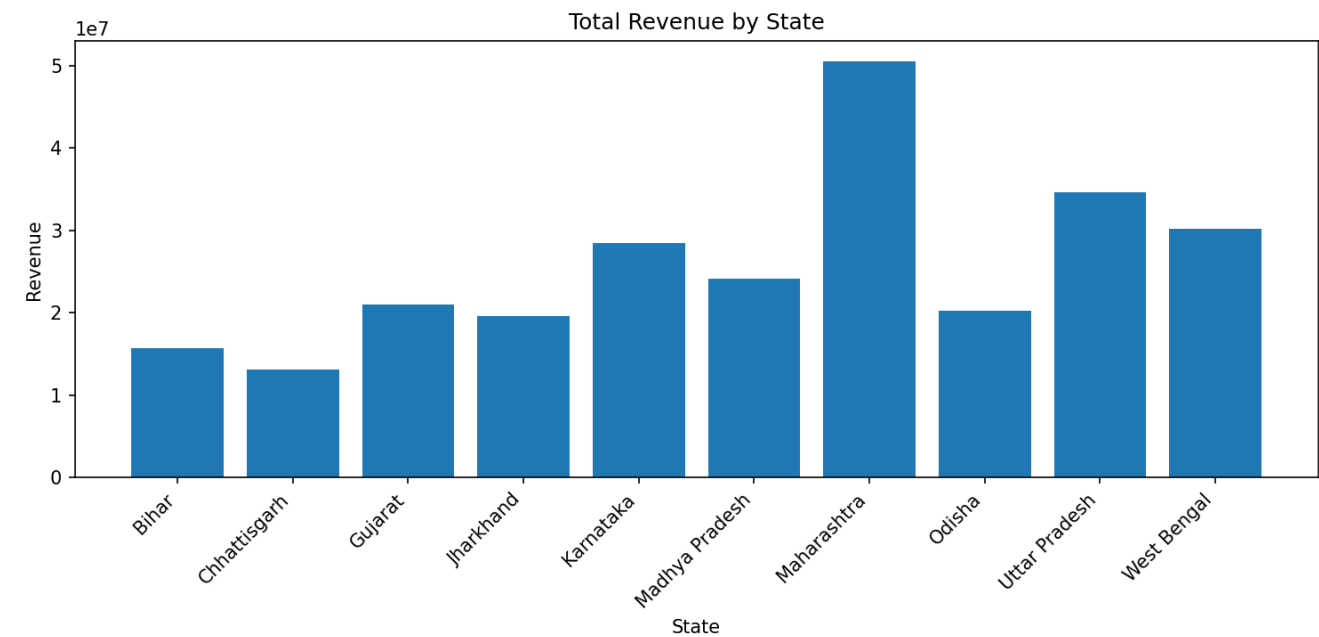
The analysis contains 30 city records across 10 states. Postal codes were treated as strings, numeric fields were converted to numeric types, and duplicate records were removed before analysis.

Key Metrics

Metric	Value
Cities analyzed	30
States represented	10
Total users	257,800
Total revenue	■257,800,000
Average revenue per user	■1,000.00

Geographic Analysis

State-level aggregation shows differences in total revenue and user density. These comparisons are descriptive of the representative dataset only.



High-Potential Region Analysis

An opportunity score was calculated using population relative to the dataset median and user penetration relative to the dataset median. Cities above the population median and at or below the penetration median were treated as candidates for further investigation. This is a demonstration heuristic, not a validated market-sizing model.

State	City	Population	Users	Penetration	Score
Maharashtra	Mumbai	12,600,000	32,000	0.25%	26.71
Karnataka	Bengaluru	8,490,000	28,500	0.34%	13.61
Gujarat	Ahmedabad	5,630,000	21,000	0.37%	8.12

Business Recommendations

- Use the highlighted cities as candidates for deeper market validation.
- Compare local competition, distribution coverage, customer acquisition cost, and demand before expansion decisions.
- Investigate whether lower user penetration reflects limited coverage, lower awareness, pricing, or other local factors.
- Refresh the analysis with official business data before using the opportunity score for operational decisions.

Interactive Deliverable

The accompanying HTML map contains city-level markers with revenue, users, population, user penetration, and opportunity score. The three highest-scoring candidate regions are highlighted.